

PawCare

CCE106 Final Project Documentation

Pet Profile and Care Schedule Management

In Partial Project Requirement of
APPLICATIONS DEV'T AND EMERGING TECHNOLOGIES
CCE106/L(3661)

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Chapter 1: Implemented Application

I. Introduction

PawCare is a Flutter mobile application that helps pet owners keep pet records and care schedules in one place. A signed-in user can create and update pet profiles, plan activities for each pet, review the day's schedules, and enable reminders. The app uses Supabase for authentication, database records, and photo storage.

The implemented version focuses on organizing pet information and planned care tasks. It provides a dashboard, pet profile management, a calendar-based schedule, and account settings. Meal suggestions and an in-app chatbot are not part of the current app interface.

II. Purpose and Value Proposition

PawCare aims to make pet care easier and more organized. Instead of depending on separate notes, phone alarms, or calendar applications, pet owners can manage pet information and care schedules in one place.

The application provides the following:

- Pet profiles for storing each pet's details, photo, health notes, and fun facts.
- A calendar-based care schedule for planning activities, setting dates and times, adding notes, and tracking completion. Users can enable local reminders for saved schedules.
- A home dashboard that shows the active pet and today's activities, with quick access to pet profiles and the schedule.

PawCare brings pet information and schedule management together in one application. Owners can review each pet's details alongside planned care activities, helping them keep routine information organized.

III. Target Audience and Pet Profiles

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PawCare is intended for pet owners who want a straightforward way to keep pet profiles and care schedules together. It supports people responsible for one or more pets and users who want to review upcoming activities from a dashboard or calendar.

IV. Core Functionalities

4.1 Pet Profile Management

Users can add, view, and edit profiles for their pets. A profile can include the pet's name, species, breed, optional birthdate, calculated age, gender, status, health note, fun fact, and photo. Users can select a profile to view its details and schedules. Pet photos are uploaded to the Supabase Storage bucket used by the app.

The My Pets screen lists the signed-in user's pet profiles. The dashboard also shows an active pet; users can select another pet or open its profile. Profile records are associated with the signed-in account.

4.2 Care Schedule

4.2 Pet Care Schedule

The Pet Schedule screen presents saved activities in a calendar and date-based list. Users can create or edit a schedule by choosing a pet, activity type, title, date, time, repeat option, and optional notes. A schedule can be marked complete or deleted. The activity is associated with its pet and appears in the schedule list and pet details.

The repeat selection is saved as schedule information. Each schedule stores its selected date and time; the current interface does not create a series of separate future records for recurring activities.

4.3 Dashboard

The Home dashboard greets the signed-in user, displays the active pet, and provides quick access to Pet Profiles and Pet Schedule. Its Today section shows schedules for the current date. Selecting the active pet opens its details.

4.4 Accounts and Profile

4.4 Account Access and Profile

PawCare provides account registration, email verification by code, email-and-password sign-in, Google sign-in, password recovery, password update, and sign-out. The authentication flow returns users to the app after supported email or OAuth callbacks.

Users can view and update their profile name and profile photo, along with account information such as their email address. The app uses Supabase Auth for account sessions.

4.5 Settings

The settings screen provides access to profile editing, notification preferences, and sign-out. Notification permission must be granted before reminders can be enabled. The app includes a separate accessibility settings screen in the source, but it is not currently linked from the main settings navigation.

4.6 Notifications and Technical Feasibility

4.6 Schedule Notifications

Users can enable or disable schedule notifications in settings. On supported platforms, PawCare requests notification permission and schedules local reminders for saved care activities. When a schedule is completed or removed, its reminder is updated or cancelled. Browser notification permission is supported, while scheduled local reminders are intended for native platforms.

V. Technical Feasibility and Tools

The application is built with Flutter and Dart. Supabase provides authentication, database access, and file storage. The main data collections used by the app are pets and pet_schedules; pet and profile photos use the pet-photos storage bucket. The project also includes a pawcare-ai Supabase Edge Function, but the current Flutter interface does not call it.

Flutter packages include image_picker and image_cropper for selecting and cropping photos, flutter_local_notifications and timezone for reminders, shared_preferences for a local notification preference, and supabase_flutter for backend access.

VI. Data and Implementation Scope

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Pet and schedule operations are performed for the signed-in user. The application filters records by owner and relies on Supabase project configuration and database policies to enforce access controls. A deployed environment needs the Supabase tables, authentication provider settings, storage bucket, and appropriate row-level security and storage policies.

The repository's SQL migration files describe incremental changes to an existing database. They are not a complete initial schema. Google sign-in also requires provider credentials and correct redirect configuration.

The implemented app organizes care information; it does not diagnose health conditions or provide veterinarian-approved treatment or nutrition advice. The current scope does not include the proposed meal recommendation experience or a user-facing chatbot. These items should be treated as future work unless they are separately implemented and connected to the app.

Chapter 1: Implemented Scope Summary

PawCare currently provides account access, pet profile management, a dashboard, care scheduling, profile settings, and schedule notifications on supported platforms. The current Flutter interface does not include meal recommendations or a chatbot.

The next chapter retains the original wireframe illustrations as design references. They show the planned dashboard, pet profile, and scheduling flows; some screen details may differ from the implemented application described in this document.

Chapter 2:

Conceptualizing the Mobile Application

VI. Wireframes

Main and Navigation Bar: The Home / Dashboard serves as the main screen of the PawCare mobile application, providing the user with an overview of the active pet, including its name, breed, and age. It also provides quick access to the My Pets and Pet Schedule sections, while the Today section displays the pet's current or upcoming schedule. The navigation bar at the bottom allows the user to easily access the main sections of the application.

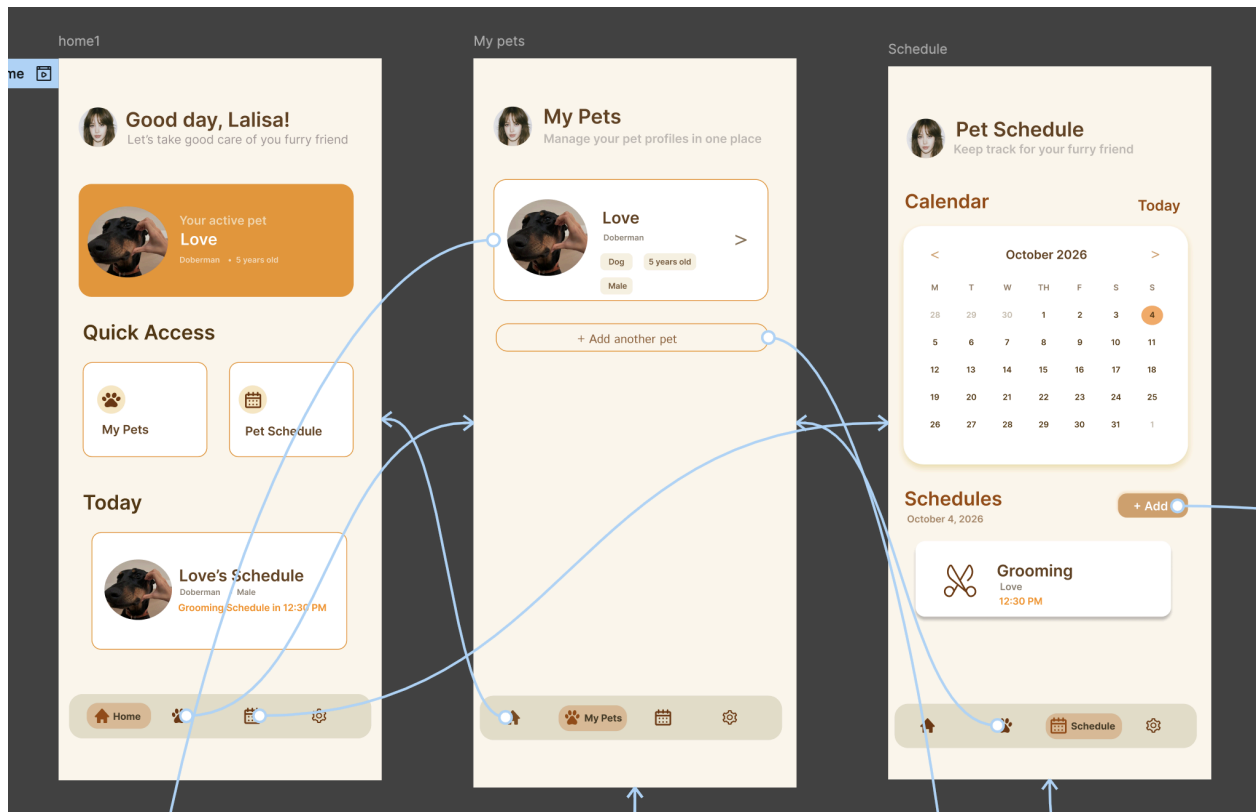


Figure 1. Home / Dashboard

My Pets: This screen allows the user to view and manage registered pet profiles in one place. Each pet is displayed with basic information such as its name, species, breed, age, and gender. The user can select a pet to view its profile or choose Add another pet to create a new pet record, while the navigation bar provides access to the other main sections of PawCare.

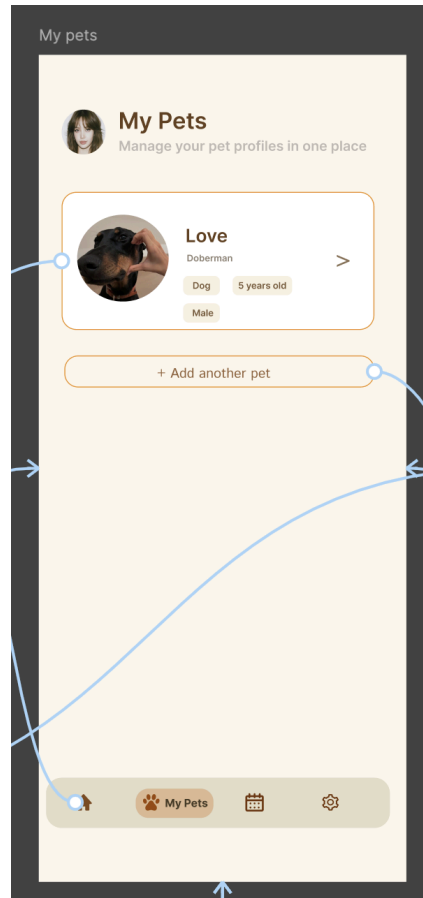


Figure 2. My Pets

Pet Profile, Edit Pet, and Add Pet: This screen allows the user to manage individual pet profiles. The Pet Profile displays the selected pet's information and scheduled activities, while the Edit Pet screen allows the user to update details such as the pet's name, species, breed, birthdate, and gender. The Add Pet screen provides the same information fields for creating a new pet profile, including an option to add a pet photo, and the user can select Save Pet to store the information

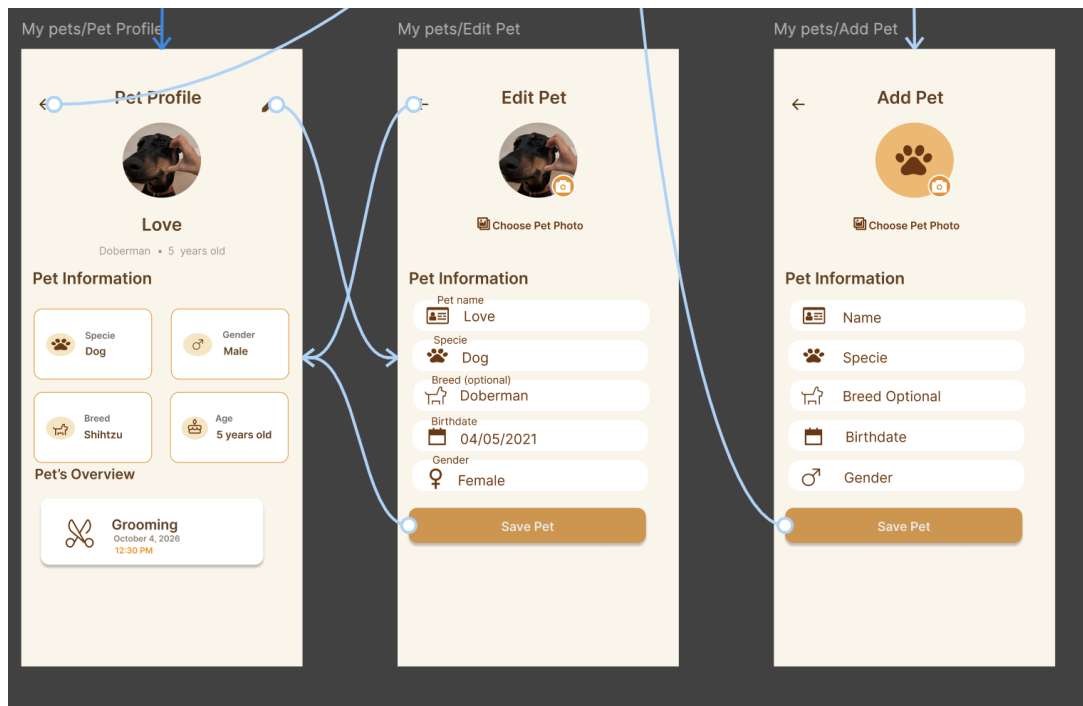


Figure 2.1 Edit Pet Information and Add Pet

Schedule: The Pet Schedule screen allows the user to view and organize their pet's activities using a calendar. After selecting a date, the schedules associated with that date are displayed below the calendar, including details such as the activity title, pet name, and scheduled time. The + Add button allows the user to create a new schedule for a selected pet.

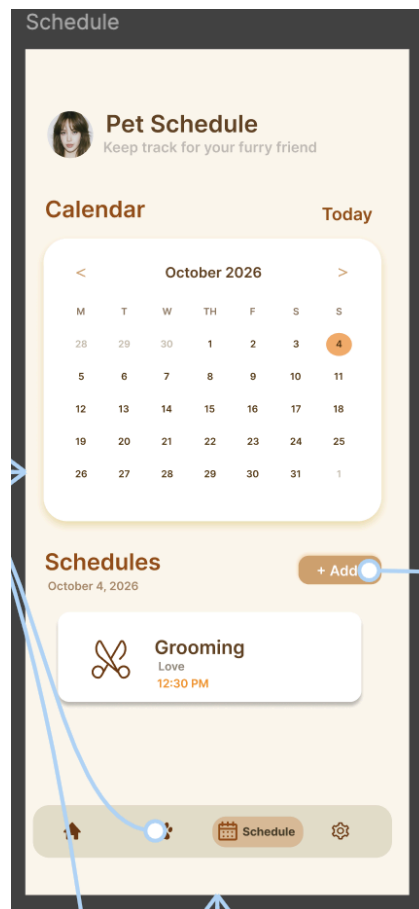


Figure 3. Schedule

Add Schedule: The Add Schedule screen allows the user to create a new activity for a pet by entering the pet name, activity type, schedule title, date, time, repeat frequency, and optional notes. After completing the required information, the user can select Save, which adds the new activity to the Pet Schedule screen where it can be viewed together with the pet's other scheduled activities.

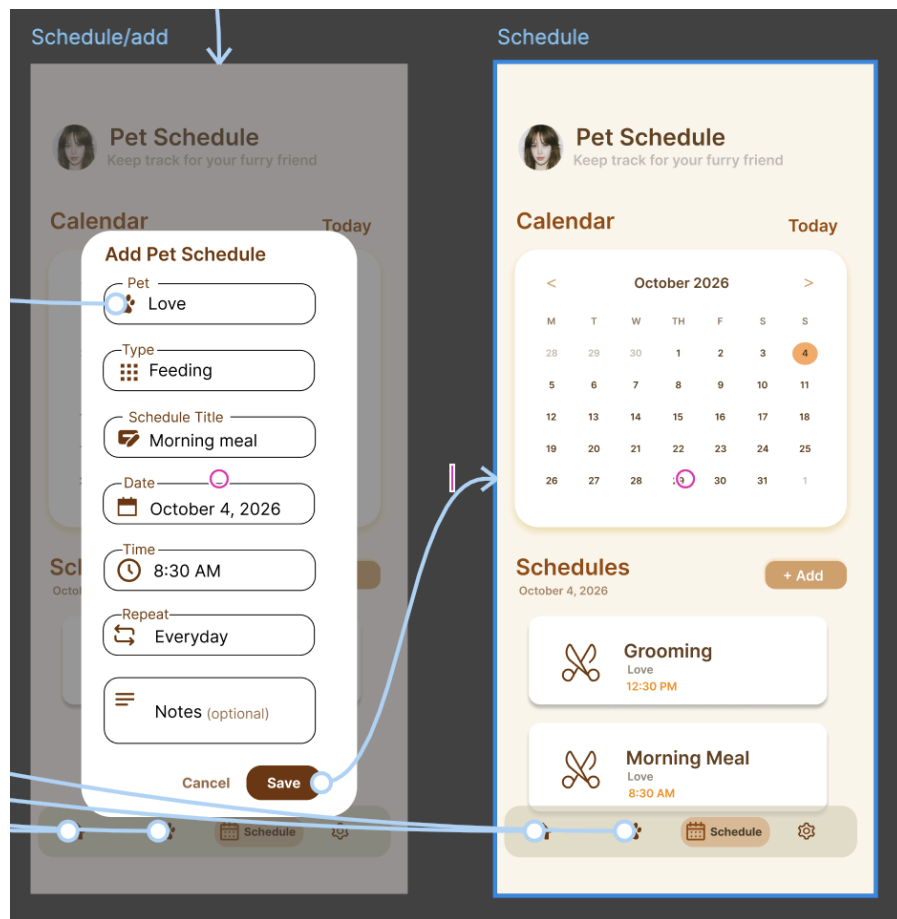


Figure 3.1. Add Schedule

VII. Task Analysis

PawCare begins at the authentication flow. A new user creates an account and verifies the email address with the code sent by Supabase. Returning users sign in with email and password or choose Google sign-in. A user who forgets a password requests a recovery code and follows the password update flow.

After sign-in, the Home dashboard shows the active pet and the day's schedules. The user can open Pet Profiles to view saved pets or add a pet by entering its details and optionally selecting a photo. Selecting a pet opens its profile, where the user can review its information and associated schedules. The edit action allows the user to change profile details or photo.

To plan care, the user opens Pet Schedule and selects a date in the calendar. The user chooses Add, selects a pet, enters an activity type and title, chooses a date and time, selects a repeat option, and may add notes. Saving adds the schedule to the calendar and date list. The user can edit or delete a schedule and mark it complete.

In settings, the user can edit profile information, turn schedule notifications on or off, and sign out. Enabling notifications prompts for device permission. On supported native platforms, saved schedules can generate local reminders.

Chapter 3: Demonstrating the Mobile Application

VIII. User Interface Manual

1. Open PawCare and sign in. New users select Sign Up, provide account details, then enter the email verification code.
2. On Home, review the active pet and today's activities. Use the bottom navigation to open Pet Profiles, Pet Schedule, or Settings.
3. In Pet Profiles, select a pet to view its profile. Use Add Pet to create a record; enter the required name, species, and gender, and optionally add other details and a photo. Save the profile. Use Edit to update an existing pet.
- 4. In Pet Schedule, select a calendar date to see that day's activities. Use Add to create a schedule: choose a pet, enter the activity type and title, set date and time, choose a repeat option, and optionally add notes. Save to add it to the schedule. Use the schedule controls to edit, complete, or delete an activity.**
5. In Settings, update profile information or enable notifications. Allow notification permission when prompted. Choose Log Out to end the session.

IX. Feature Highlights

Pet profiles keep each pet's identifying details, notes, and photo together. The date-based care schedule links activities to a selected pet and supports editing and completion tracking. The dashboard provides a quick view of the active pet and today's schedules. Account authentication keeps each user's app session distinct, while local reminders help users notice planned care activities when notification permissions are enabled.

Current implementation note: Meal recommendations and an in-app chatbot are not exposed in the Flutter app. The included pawcare-ai Edge Function is not called by the current UI.